# Vector autoregression II

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# Plan for this slide deck

1. How to convert a VAR($p$) model to VAR(1) so we can...
   - Check stationarity
   - Forecast
   - Compute moments
2. VARMA model, matrix lag polynomials
3. VAR bootstrap for more accurate small-sample inference

# VAR($p$): companion form

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}_1 \boldsymbol{y}_{t-1} + \cdots + \boldsymbol{A}_p \boldsymbol{y}_{t-p} + \boldsymbol{u}_t$$

- When is the model cov. stationary? How do we forecast? What are the moments?
- Everything will follow from our previous VAR(1) results after we perform a neat trick...

# VAR($p$): companion form (cont.)

- Claim: the VAR($p$) model with $n$ variables is *equivalent* with the following **companion form** VAR(1) model with $n \times p$ variables:

$$\tilde{\boldsymbol{y}}_t = \tilde{\boldsymbol{c}} + \tilde{\boldsymbol{A}}\tilde{\boldsymbol{y}}_{t-1} + \tilde{\boldsymbol{u}}_t, \quad \text{where}$$

$$\tilde{\boldsymbol{y}}_t \equiv \underbrace{\begin{pmatrix} \boldsymbol{y}_t \\ \boldsymbol{y}_{t-1} \\ \vdots \\ \boldsymbol{y}_{t-p+1} \end{pmatrix}}_{np\times 1}, \quad \tilde{\boldsymbol{c}} \equiv \underbrace{\begin{pmatrix} \boldsymbol{c} \\ \boldsymbol{0} \\ \vdots \\ \boldsymbol{0} \end{pmatrix}}_{np\times 1}, \quad \tilde{\boldsymbol{u}}_t \equiv \underbrace{\begin{pmatrix} \boldsymbol{u}_t \\ \boldsymbol{0} \\ \vdots \\ \boldsymbol{0} \end{pmatrix}}_{np\times 1},$$

$$\tilde{\boldsymbol{A}} \equiv \underbrace{\left(\begin{array}{ccc|c} \boldsymbol{A}_1 & \cdots & \boldsymbol{A}_{p-1} & \boldsymbol{A}_p \\ \hline \multicolumn{3}{c|}{\boldsymbol{I}_{n(p-1)}} & \boldsymbol{0}_{n(p-1)\times n} \end{array}\right)}_{np\times np}.$$

# VAR($p$): companion form (cont.)

- Example: consider a univariate AR(2) model

$$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + u_t.$$

- Companion form:

$$\begin{pmatrix} y_t \\ y_{t-1} \end{pmatrix} = \begin{pmatrix} c \\ 0 \end{pmatrix} + \begin{pmatrix} a_1 & a_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} y_{t-1} \\ y_{t-2} \end{pmatrix} + \begin{pmatrix} u_t \\ 0 \end{pmatrix}.$$

  Only the first equation is non-vacuous.

- Exercise: do the same for an AR(3) model.

# VAR($p$): companion form (cont.)

- When does a VAR($p$) model have a cov. stationary solution?
  - Answer: when the companion form VAR matrix $\tilde{\boldsymbol{A}}$ has all its eigenvalues inside the unit circle.
- How can we forecast (one-step or multi-step) with a VAR($p$) model?
  - Answer: rewrite model in companion form, then apply the simple VAR(1) formulas.
- What are the moments of the variables in a VAR($p$) model?
  - Answer: write model in companion form and use the VAR(1) formulas.

# VARMA models

- Vector ARMA (**VARMA**) model:

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}_1\boldsymbol{y}_{t-1} + \cdots + \boldsymbol{A}_p\boldsymbol{y}_{t-p} + \boldsymbol{u}_t + \boldsymbol{\Theta}_1\boldsymbol{u}_{t-1} + \cdots + \boldsymbol{\Theta}_q\boldsymbol{u}_{t-q},$$

  where $\boldsymbol{\Theta}_1, \ldots, \boldsymbol{\Theta}_q$ are $n \times n$ matrices.

- Can write this with **matrix lag polynomial** notation:

$$\boldsymbol{A}(L)\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{\Theta}(L)\boldsymbol{u}_t, \quad \text{where}$$

$$\boldsymbol{A}(L) \equiv \boldsymbol{I} - \sum_{\ell=1}^{p} \boldsymbol{A}_\ell L^\ell, \quad \boldsymbol{\Theta}(L) \equiv \boldsymbol{I} + \sum_{\ell=1}^{q} \boldsymbol{\Theta}_\ell L^\ell.$$

- Matrix lag polynomials work like scalar ones, except they don't necessarily commute (like ordinary matrices): $\boldsymbol{B}(L)\boldsymbol{C}(L) \neq \boldsymbol{C}(L)\boldsymbol{B}(L)$.

# Finite-sample properties of VAR estimator

- The OLS estimator for VAR($p$) models is approximately unbiased and normally distributed when the sample size $T$ is large.
- But as we saw for AR models, when $T$ is small/moderate and the data is very persistent, the estimator could be quite biased and non-normally distributed.
- As a result, the standard OLS confidence intervals may not have coverage probability close to the nominal level.

# Bootstrap

- The **bootstrap** is a method for obtaining a better approximation to the small-sample distribution of the VAR estimator than that provided by asymptotic theory.
- Instead of theoretical approximations, the bootstrap leverages computing power.
- The basic idea is to **resample** the data many times:
  - Generate 100s of simulated data sets that mimic the actual data generating process.
  - Compute the VAR estimator on each of the simulated data sets.
  - This traces out a sampling distribution for the VAR estimator across *simulated* data sets.
  - Hopefully provides a good approximation to the sampling distribution across *hypothetical histories of the real world*.

# Bootstrap: motivation

- Consider a VAR(1) model with i.i.d. innovations:

$$\boldsymbol{y}_t = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1} + \boldsymbol{u}_t, \quad \boldsymbol{u}_t \overset{iid}{\sim} (\boldsymbol{0}, \boldsymbol{\Sigma}).$$

  - Could be companion form of a VAR($p$) model.

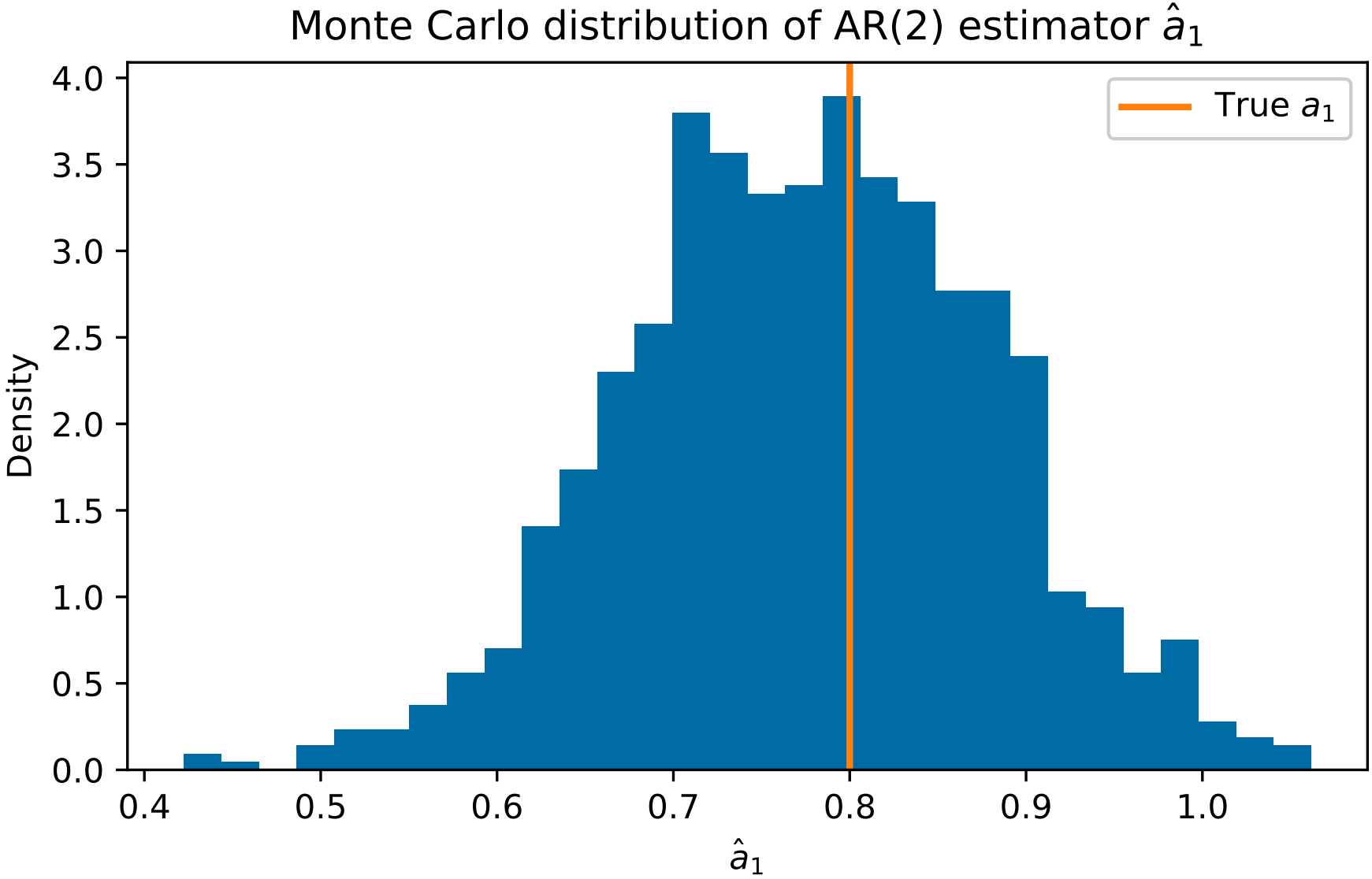
- We want to understand the bias and distributional properties of the OLS estimator $\hat{\boldsymbol{A}}$ in a given sample size $T$.

- Suppose for a moment that we *knew* the true VAR coefficients $\boldsymbol{c}, \boldsymbol{A}$ and the true distribution of $\boldsymbol{u}_t$ (a bit silly, but stay with me!).

- Then the most straight-forward way to find the distribution of $\hat{\boldsymbol{A}}$ is through **Monte Carlo simulation** (we already did this in an earlier slide deck).

# Monte Carlo simulation: algorithm

1. Choose a large $B$ (e.g., 1000).
2. In each **iteration** $b = 1, 2, \ldots, B$:
   a) Simulate shocks $\boldsymbol{u}_2^*, \ldots, \boldsymbol{u}_T^*$ from the known distribution of $\{\boldsymbol{u}_t\}$.
   b) Generate a fake data set $\{\boldsymbol{y}_t^*\}_{t=1}^T$ satisfying the known model:
   $$\boldsymbol{y}_t^* = \boldsymbol{c} + \boldsymbol{A}\boldsymbol{y}_{t-1}^* + \boldsymbol{u}_t^*, \quad t = 2, \ldots, T,$$
   given an initial value $\boldsymbol{y}_1^*$ drawn from a known distribution.
   c) Compute the VAR OLS estimator $\hat{\boldsymbol{A}}_{(b)}^*$ on the fake sample $\{\boldsymbol{y}_1^*, \ldots, \boldsymbol{y}_T^*\}$.
3. The histogram of the $B$ simulated estimates $\hat{\boldsymbol{A}}_{(1)}^*, \ldots, \hat{\boldsymbol{A}}_{(B)}^*$ approximates the actual sampling distribution of the VAR OLS estimator for sample size $T$.




$$y_t = c + a_1 y_{t-1} + a_2 y_{t-2} + u_t, \quad u_t \overset{iid}{\sim} N(0, \sigma^2),$$

$$c = 0,\ a_1 = 0.8,\ a_2 = 0.1,\ \sigma^2 = 1, \quad T = 100, \quad B = 1000$$

# Recursive residual bootstrap

- The preceding strategy is infeasible in practice, since we know neither the model coefficients, nor the shock distribution.
- But we can *estimate* these:
  - Approximate the true $\boldsymbol{c}, \boldsymbol{A}$ with the real-data estimates $\hat{\boldsymbol{c}}, \hat{\boldsymbol{A}}$.
  - Approximate the shock distribution by the empirical distribution of the OLS residuals $\hat{\boldsymbol{u}}_2, \ldots, \hat{\boldsymbol{u}}_T$.
- Plugging these estimates of the true parameters into the previous Monte Carlo recipe yields the **recursive residual bootstrap**.

# Recursive residual bootstrap: algorithm

1. Compute real-data OLS estimates $\hat{\boldsymbol{c}}, \hat{\boldsymbol{A}}$ and residuals $\hat{\boldsymbol{u}}_2, \ldots, \hat{\boldsymbol{u}}_T$.
2. Choose a large $B$ (e.g., 1000).
3. In each iteration $b = 1, 2, \ldots, B$:
   a) Draw $T-1$ shock vectors $\boldsymbol{u}_2^*, \ldots, \boldsymbol{u}_T^*$ uniformly at random (with replacement) from the list $\hat{\boldsymbol{u}}_2, \ldots, \hat{\boldsymbol{u}}_T$.
   b) Generate a fake data set $\{\boldsymbol{y}_t^*\}_{t=1}^T$ recursively:
   $$\boldsymbol{y}_t^* = \hat{\boldsymbol{c}} + \hat{\boldsymbol{A}}\boldsymbol{y}_{t-1}^* + \boldsymbol{u}_t^*, \quad t = 2, 3, \ldots, T,$$
   given an initial value $\boldsymbol{y}_1^*$ (more on next slide).
   c) Compute the VAR OLS estimator $\hat{\boldsymbol{A}}_{(b)}^*$ on the fake sample $\{\boldsymbol{y}_1^*, \ldots, \boldsymbol{y}_T^*\}$.
4. The histogram of the $B$ bootstrap estimates $\hat{\boldsymbol{A}}_{(1)}^*, \ldots, \hat{\boldsymbol{A}}_{(B)}^*$ approximates the sampling distribution of $\hat{\boldsymbol{A}}$ in the real data.

# Recursive residual bootstrap: initial values

- When simulating bootstrap data $\{\boldsymbol{y}_t^*\}$, we need to pick an initial value $\boldsymbol{y}_1^*$.
- It's customary to pick this value uniformly at random from the real data set, independently in each bootstrap iteration.
- VAR($p$) generalization: need $p$ initial values. We can pick $p$ consecutive observations uniformly at random from the real data as initial values.

# Recursive residual bootstrap: example

- The VAR(1) algorithm easily generalizes to VAR($p$).
- Let's do an AR(2) example for concreteness.

```
# Load data
df = pd.read_csv('data/CPIAUCSL.csv', index_col='observation_date',
    parse_dates=True)
df = df.interpolate() # interpolate missing value in Oct 2025
cpi = df['CPIAUCSL'].values
y = 100*np.diff(np.log(cpi)) # monthly inflation rate

# Estimate AR(2) model
result = sm.tsa.AutoReg(y, lags=2).fit()
print(result.summary())
```

AutoReg Model Results

| | | | |
|---|---|---|---|
| Dep. Variable: | y | No. Observations: | 949 |
| Model: | AutoReg(2) | Log Likelihood | -113.499 |
| Method: | Conditional MLE | S.D. of innovations | 0.273 |
| Date: | Wed, 08 Apr 2026 | AIC | 234.999 |
| Time: | 09:11:22 | BIC | 254.412 |
| Sample: | 2 | HQIC | 242.397 |
| | 949 | | |

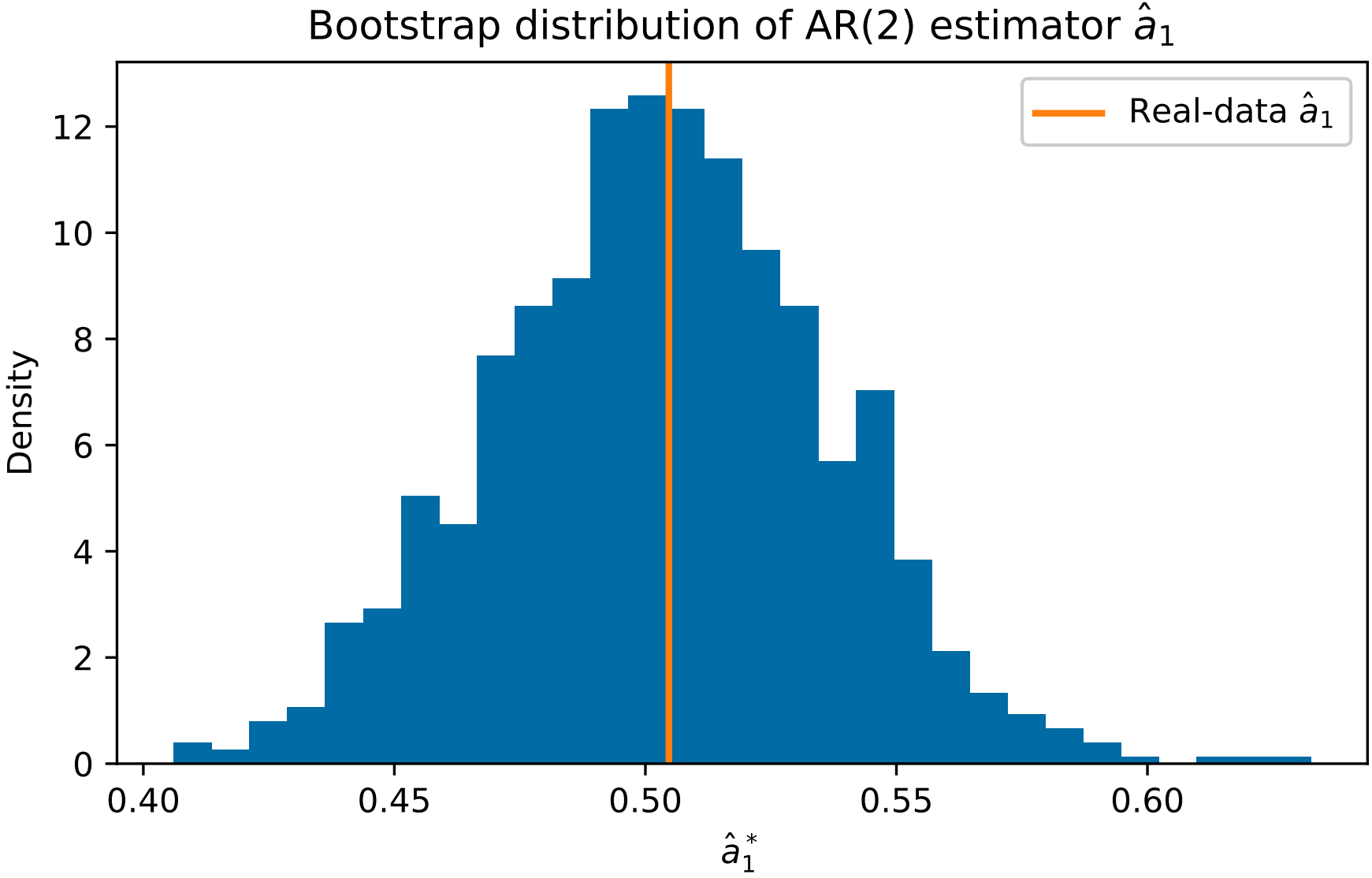
| | coef | std err | z | P>\|z\| | [0.025 | 0.975] |
|---|---|---|---|---|---|---|
| const | 0.1081 | 0.012 | 8.823 | 0.000 | 0.084 | 0.132 |
| y.L1 | 0.5047 | 0.032 | 15.802 | 0.000 | 0.442 | 0.567 |
| y.L2 | 0.1126 | 0.032 | 3.529 | 0.000 | 0.050 | 0.175 |

```
param_hat = result.params
u_hat = result.resid # residuals (length T-2)
T = len(y)

B = 1000 # number of bootstrap samples

param_hat_boot = np.empty((B, 3))
for b in range(B):
    # Resample residuals with replacement
    u_boot = rng.choice(u_hat, size=T - 2, replace=True)
    # Initial values: random pair of consecutive obs.
    i0 = rng.integers(0, T - 1)
    y_init = y[i0:i0 + 2]
    # Simulate bootstrap sample and estimate
    y_boot = simulate_ar2(param_hat, u_boot, y_init)
    result_boot = sm.tsa.AutoReg(y_boot, lags=2).fit()
    param_hat_boot[b] = result_boot.params
```

Note: the `simulate_ar2` function is defined by me (not shown here).


Bootstrap distribution of AR(2) estimator $\hat{a}_1$
Real-data $\hat{a}_1$
Density
0
2
4
6
8
10
12
0.40
0.45
0.50
0.55
0.60
$\hat{a}_1^*$

# Bootstrap approximation of sampling distribution

- The coefficients that generate the bootstrap data sets are $\hat{\boldsymbol{A}}$.
- Hence, the distribution of $\hat{\boldsymbol{A}}^* - \hat{\boldsymbol{A}}$ across bootstrap iterations is analogous to the distribution of $\hat{\boldsymbol{A}} - \boldsymbol{A}$ across (hypothetical) repeated samples of the real data.
  - $\boldsymbol{A}$: true coefficients that generated the real data.
- Result: the bootstrap distribution of $\hat{\boldsymbol{A}}^* - \hat{\boldsymbol{A}}$ approximates the sampling distribution of $\hat{\boldsymbol{A}} - \boldsymbol{A}$ arbitrarily well as $T \to \infty$.
  - Assumes the data follows a stationary VAR(1) model with i.i.d. shocks.
- In practice for small/moderate $T$, the bootstrap approximation is typically more accurate than the approximation provided by our usual analytical asymptotic formulas (used in regression software).

# Bootstrap standard errors

- We would like to report standard errors (SEs) for our VAR estimator. SE = estimate of the standard deviation of an element $\hat{A}_{ij}$ of the matrix $\hat{\boldsymbol{A}}$.
- We can use the bootstrap distribution to compute SEs.
- The **bootstrap standard error** of the estimator $\hat{A}_{ij}$ is the sample standard deviation of $\hat{A}^*_{ij}$ across the $B$ bootstrap draws.

```
# Bootstrap SE
se_boot = np.std(param_hat_boot, axis=0, ddof=1)
```

# Bootstrap confidence interval

- We want a confidence interval (CI) for $A_{ij}$.
- Let $\tilde{q}_\alpha$ be the $\alpha$-quantile of the sampling distribution of $\hat{A}_{ij} - A_{ij}$ in the real data. By definition,

$$P\left(\tilde{q}_{\alpha/2} \leq \hat{A}_{ij} - A_{ij} \leq \tilde{q}_{1-\alpha/2}\right) = 1 - \alpha.$$

- Let $\tilde{q}^*_\alpha$ be the $\alpha$-quantile of $\hat{A}^*_{ij} - \hat{A}_{ij}$ across the $B$ bootstrap draws (i.e., the $(\alpha \times B)$-th smallest value).
- Because the bootstrap distribution of $\hat{A}^*_{ij} - \hat{A}_{ij}$ approximates the sampling distribution of $\hat{A}_{ij} - A_{ij}$,

$$\tilde{q}^*_\alpha \approx \tilde{q}_\alpha \quad \text{for all } \alpha$$

$$\implies \quad P\left(\tilde{q}^*_{\alpha/2} \leq \hat{A}_{ij} - A_{ij} \leq \tilde{q}^*_{1-\alpha/2}\right) \approx 1 - \alpha.$$

# Bootstrap confidence interval (cont.)

$$P\left(\tilde{q}^*_{\alpha/2} \leq \hat{A}_{ij} - A_{ij} \leq \tilde{q}^*_{1-\alpha/2}\right) \approx 1 - \alpha$$

- Rearrange the inequalities:

$$P\left(\hat{A}_{ij} - \tilde{q}^*_{1-\alpha/2} \leq A_{ij} \leq \hat{A}_{ij} - \tilde{q}^*_{\alpha/2}\right) \approx 1 - \alpha.$$

- Conclusion: the **bootstrap confidence interval**

$$\hat{\mathcal{I}}^* \equiv \left[\hat{A}_{ij} - \tilde{q}^*_{1-\alpha/2}, \hat{A}_{ij} - \tilde{q}^*_{\alpha/2}\right]$$

has approximately correct coverage:

$$\boxed{P\left(A_{ij} \in \hat{\mathcal{I}}^*\right) \approx 1 - \alpha}.$$

(This becomes an equality in the limit $T \to \infty$.)

# Bootstrap confidence interval: alternative formula

- Since $\hat{A}_{ij}$ doesn't change across bootstrap iterations (only $\hat{A}^*_{ij}$ does),

$$\tilde{q}^*_\alpha = q^*_\alpha - \hat{A}_{ij},$$

  where $q^*_\alpha$ is the $\alpha$-th sample quantile of $\hat{A}^*_{ij}$ across bootstrap draws.

- Hence, we can equivalently write the bootstrap CI as

$$\hat{\mathcal{I}}^* = \left[2\hat{A}_{ij} - q^*_{1-\alpha/2}, 2\hat{A}_{ij} - q^*_{\alpha/2}\right].$$

  - Note: the $1-\alpha/2$ quantile appears in the *left* end point of the interval, while the $\alpha/2$ quantile appears in the *right* end point.

```
# Bootstrap confidence intervals
signif = 0.05
ci_lower = 2*param_hat - np.quantile(param_hat_boot, 1 - signif/2, axis=0)
ci_upper = 2*param_hat - np.quantile(param_hat_boot, signif/2, axis=0)
```

| | Estimate | SE | CI lower | CI upper |
|---|---|---|---|---|
| const | 0.1081 | 0.0125 | 0.0810 | 0.1302 |
| a1 | 0.5047 | 0.0331 | 0.4430 | 0.5700 |
| a2 | 0.1126 | 0.0317 | 0.0532 | 0.1811 |

- In this case, the bootstrap SEs and CIs are close to the analytical ones shown on an earlier slide.
- This is because the data is only moderately persistent and $T$ is fairly large.
- In other data sets, the bootstrap can give very different results from analytical inference.

# Bootstrap confidence interval: implicit bias correction

$$\hat{\mathcal{I}}^* = \left[2\hat{A}_{ij} - q^*_{1-\alpha/2}, 2\hat{A}_{ij} - q^*_{\alpha/2}\right]$$

- The bootstrap CI may not be symmetric around $\hat{A}_{ij}$.
- If the bootstrap draws $\hat{A}^*_{ij}$ tend to be below $\hat{A}_{ij}$ (indicating downward bias), then $\hat{\mathcal{I}}^*$ is centered *above* $\hat{A}_{ij}$ – implicitly correcting the bias! (Though not perfectly.)
- Compare with the usual analytical confidence interval:

$$\hat{\mathcal{I}} \equiv \left[\hat{A}_{ij} - SE(\hat{A}_{ij})Q_{1-\alpha/2}, \hat{A}_{ij} + SE(\hat{A}_{ij})Q_{1-\alpha/2}\right],$$

where $Q_{1-\alpha/2}$ is the $(1-\alpha/2)$-quantile of the standard normal distribution. This CI is symmetric around the (biased) estimator $\hat{A}_{ij}$.

# Efron's bootstrap confidence interval

- Suppose it happened to be the case that the bootstrap distribution of $\hat{A}^*_{ij}$ was exactly symmetric around $\hat{A}_{ij}$. Then

$$q^*_{1-\alpha/2} - \hat{A}_{ij} = \hat{A}_{ij} - q^*_{\alpha/2}.$$

- Then the bootstrap CI would equal

$$\hat{\mathcal{I}}^* = \left[q^*_{\alpha/2}, q^*_{1-\alpha/2}\right].$$

- The above CI formula is called the **Efron bootstrap confidence interval**. It's popular in practice, but typically has less accurate coverage than our earlier CI, unless the estimator is nearly symmetrically distributed around the true parameter.

# Bootstrap: extensions

1. In my example code, I've worked with an AR(2) model. In the pset, you will work with a more general VAR($p$) model.
2. Suppose we want to do inference on any function $g(\boldsymbol{A}, \Sigma)$ of the VAR parameters.
   - Bootstrap SE: sample standard deviation of $g(\hat{\boldsymbol{A}}^*, \hat{\Sigma}^*)$ across the bootstrap draws.
   - Bootstrap CI: $\left[2g(\hat{\boldsymbol{A}}, \hat{\Sigma}) - q^*_{1-\alpha/2}, 2g(\hat{\boldsymbol{A}}, \hat{\Sigma}) - q^*_{\alpha/2}\right]$, where now $q^*_\alpha$ is the sample quantile of $g(\hat{\boldsymbol{A}}^*, \hat{\Sigma}^*)$ across bootstrap draws.
3. The recursive residual bootstrap assumes that the shocks are i.i.d. and therefore homoskedastic. One can allow for heteroskedasticity using a "wild bootstrap" or "block bootstrap".
   - Details: Kilian & Lütkepohl (2017), chapter 12.2, on reading list.

# What did we learn?

- VAR($p$) model: convert to VAR(1) via companion form
- VARMA model, matrix lag polynomials
- Bootstrap
  - Standard errors
  - Confidence interval

# Looking ahead

- If we run general regressions (not necessarily in the context of a VAR model) on time series data, how can we compute SE?